Saying less, steering more

Simulation 2 (asynchronous activation), setup `task003-nosolution`: first results

LLM-free simulator; extended with the efficiency of control, 12 October 2026; runs of 9 October 2026

How the simulator works. The world, `task_003`, has three candidate allocations; A0 is the correct one. Each of 24 agents is given one true fact about it and reasons as an exact Bayesian: its *belief* is the exact probability of A0, A1 and A2 given the facts it currently remembers. There are no rounds. Each agent acts at random times, on average once per time unit: it reads q posts drawn at random from the 48 most recent posts on a shared *board* (never its own), adds their facts to its memory, recomputes its belief, *votes* by probability matching (a vote drawn from its belief) and posts one remembered fact that supports its vote, or nothing if none does. Every remembered fact is forgotten after a random lifetime (on average a fraction ρ survives one time unit; reading it again renews it). A *controller* with a target, A0 (truth) or A2 (false), acts at random times at rate λ_c until $t = 30$: it reads q_c recent agent posts and, unless the *vote gate* (at least 75% of them from agents voting for its target) or the *proof stop* (its target already proved) silences it, posts the one fact from its pool that best supports its target, at most 30 messages per episode. Agents continue alone until $t = 40$. Every episode is run three times with the same random draws: without a controller, with the truth controller and with the false controller.

Summary. 24 agents hold evidence that, pooled, is a coin flip between allocation A0 (the truth) and A2, with no proof available to anyone. Simulation 2 removes the day/night rounds of Simulation 1: agents and the controller act at random times, remembered facts fade continuously, and the shared board fades too. Both controllers still tip the swarm, and **the effect persists** after they stop. When agents forget, control is **weaker than in Simulation 1** (truth 0.29 against 0.42; false 0.19 against 0.28); with full memory it is the same. The controller's rate barely matters when agents forget. With full memory, **a faster controller steers less**: it puts more of its pool into circulation, and the whole pool pulls every belief toward the same point whatever the target. Turning the vote gate off adds about 16 messages per episode and no control. Main hypotheses were confirmed on a second, independent set of random draws.

1 Setup

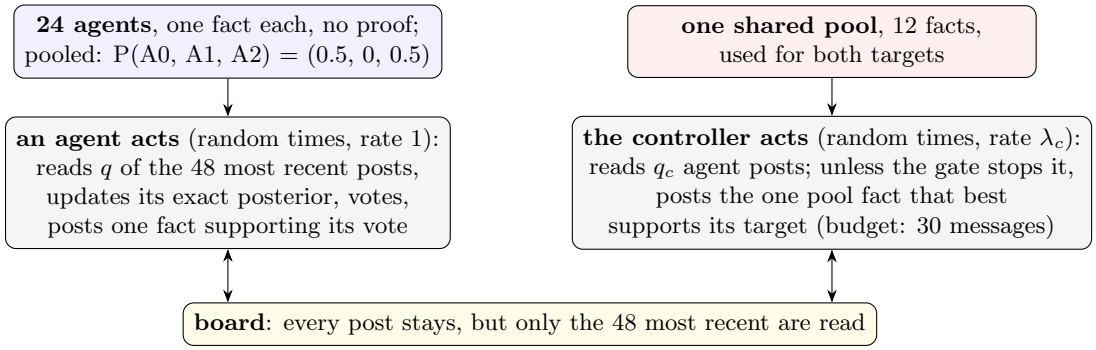


Diagram 1. Simulation 2. There are no rounds. Each remembered fact is forgotten after a random lifetime (on average a fraction ρ survives one time unit) and comes back if read again. The controller acts for $0 \le t < 30$; agents continue alone until $t = 40$.

The agents' facts plus the *whole* pool give P(A0, A1, A2) = (0.75, 0, 0.25): the truth keeps an edge that cannot be designed away, because every fact is true. Agents are exact Bayesian reasoners and vote by probability matching (the vote is drawn from the posterior). The *vote gate* keeps the controller silent when at least 75% of the posts it read come from agents who voted for its target. The grid: posts read $q \in \{3, 6, 12\}$, $q_c \in \{3, 6, 12, 24\}$, memory $\rho \in \{0.75, 1\}$, controller rate $\lambda_c \in \{0.5, 1, 2, 4, 8\}$ actions per time unit, gate on/off; 1,000 episodes per setting. Control is measured by the *gain*

$$G(t) = \frac{1}{E}\sum_{e=1}^{E}\Big[m_{\text{target}}(t)^{\text{ctrl}}(e) - m_{\text{target}}(t)^{\text{silent}}(e)\Big], \tag{1}$$

where $m_k(t)$ is the agents' mean belief in allocation k and each episode e uses the same random draws with and without the controller. Unless stated otherwise: $q = 6$, $q_c = 12$, gate on.

2 Main result: both controllers tip the swarm, and the effect stays

Table 1: Gain at $t = 30$, Eq. (1), $\lambda_c = 1$, gate on; in brackets the silent swarm's belief in the target. Simulation 1: one fact per night, same q, q_c, ρ.

ρ	controller	Simulation 2	Simulation 1	messages sent
0.75	truth (A0)	0.29 (0.38)	0.42	11
0.75	false (A2)	0.19 (0.48)	0.28	12
1	truth (A0)	0.38 (0.50)	0.38	7
1	false (A2)	0.14 (0.50)	0.15	15

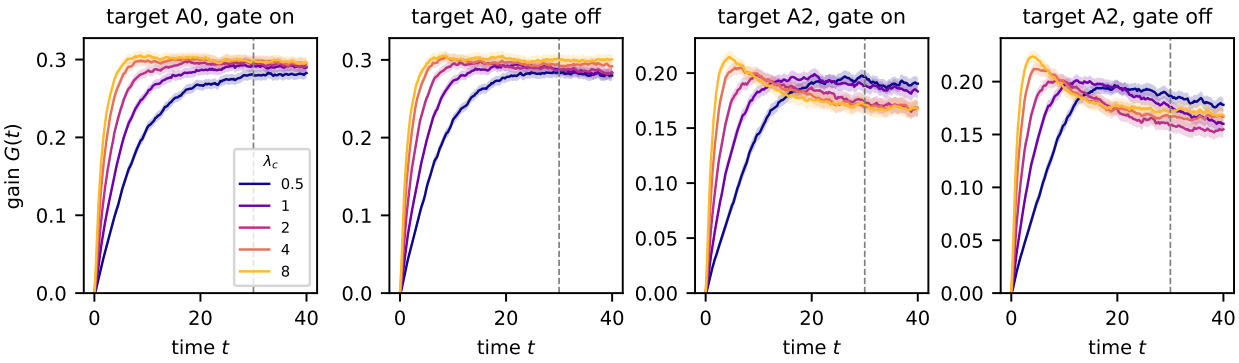


Figure 1: Gain over time by controller rate λ_c, $\rho = 0.75$; the controller stops at $t = 30$ (dashed). Bands: 95% intervals.

- **Both controllers tip the swarm** (Table 1); the truth controller is stronger, as in Simulation 1.
- **When agents forget, control is weaker than in Simulation 1**, by 0.14 (truth) and 0.09 (false) with fresh random draws. With full memory the two simulations agree within 0.01. In Simulation 1 memories change only at dawn and every agent acts right after a full day of posts; here facts expire at random moments between actions, so fewer of the controller's facts are held together at any time.
- **The effect persists**: ten time units after the controller stops, the gain has changed by at most 0.016 at $\rho = 0.75$ (Figure 1); agents keep relaying the controller's facts (12–45% of their posts).

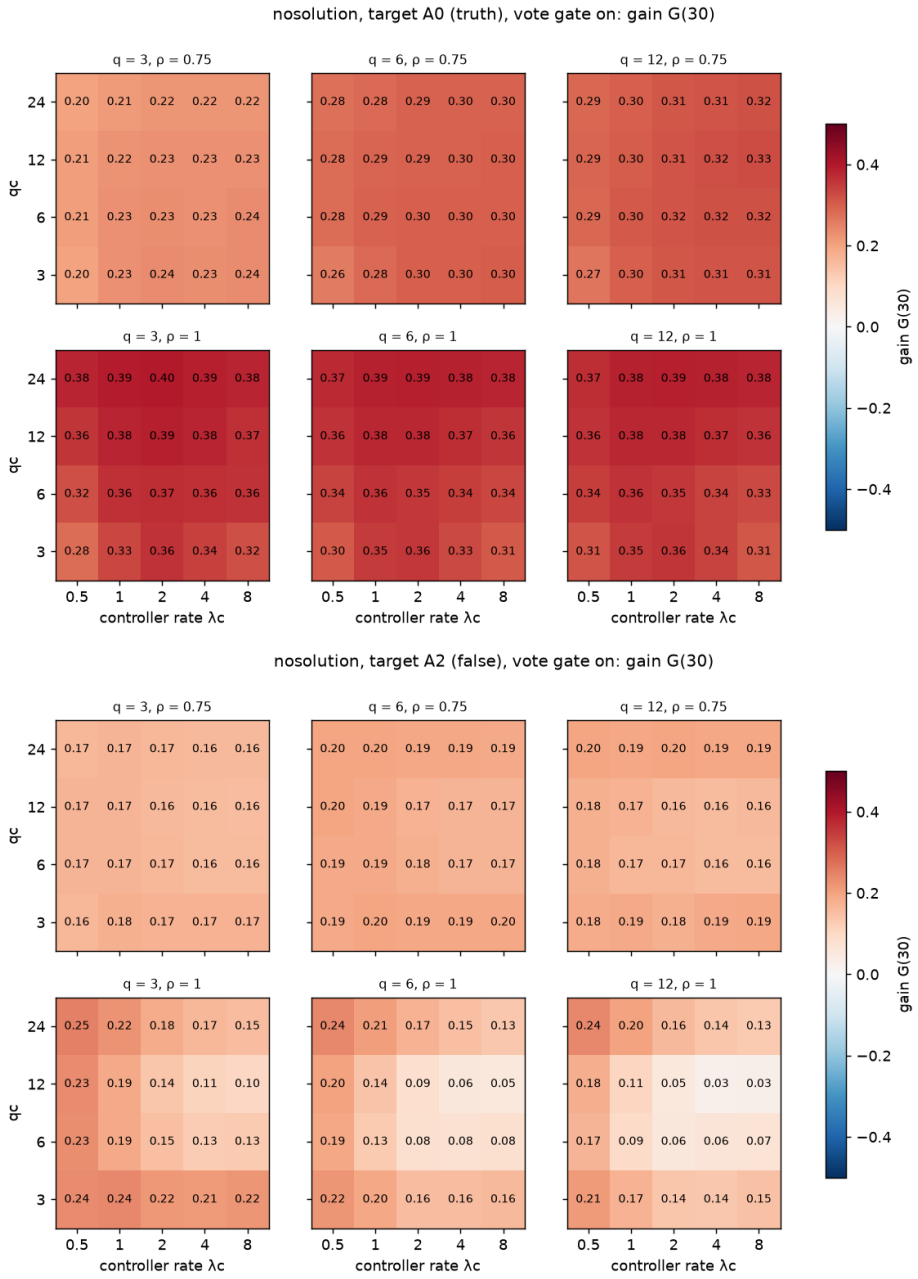


Figure 2: Gain $G(30)$ over controller rate (x) and posts read by the controller q_c (y), for every q (columns) and ρ (rows); gate on. Top: target A0; bottom: target A2.

3 The controller's rate

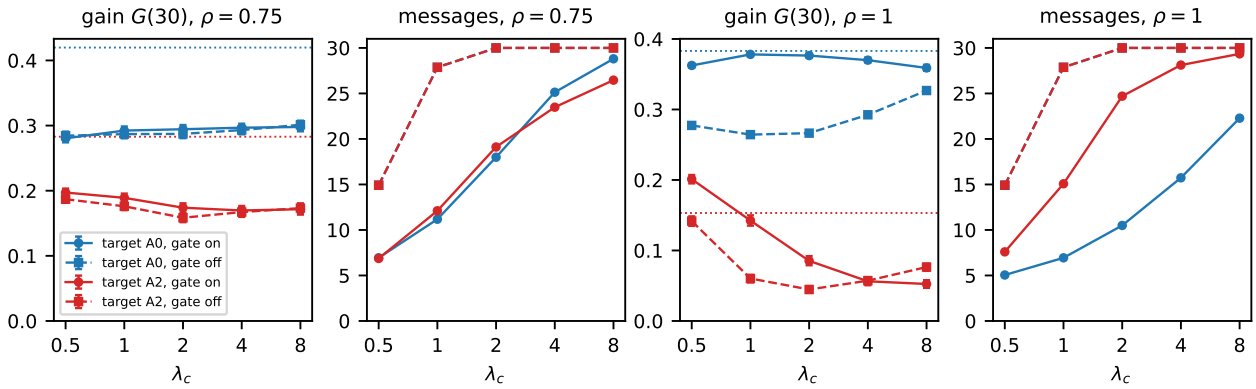


Figure 3: Gain at $t = 30$ and messages sent against the controller rate, for $\rho = 0.75$ and $\rho = 1$ (target A0 blue, A2 red; gate on solid, off dashed; dotted: Simulation 1).

When agents forget, the rate barely matters. From $\lambda_c = 0.5$ to 8 the gain moves from 0.28 to 0.30 (truth) and from 0.20 to 0.17 (false), but the messages sent rise from about 7 to 26–29. A fast controller spends its budget early (by $t \approx 4$ with the gate off), reaches its gain sooner, and then the gain stays put. Gain per message is highest for a slow, gated controller.

With full memory, a faster controller steers less. The false controller's gain falls from 0.20 at $\lambda_c = 0.5$ to 0.05 at $\lambda_c = 8$, and with the gate off the truth controller loses 0.05 on average against the gate on, up to 0.12 (Figure 3, right). The same fall appears at every q and at $q_c = 3$, 6 and 12 (Figure 2, bottom rows).

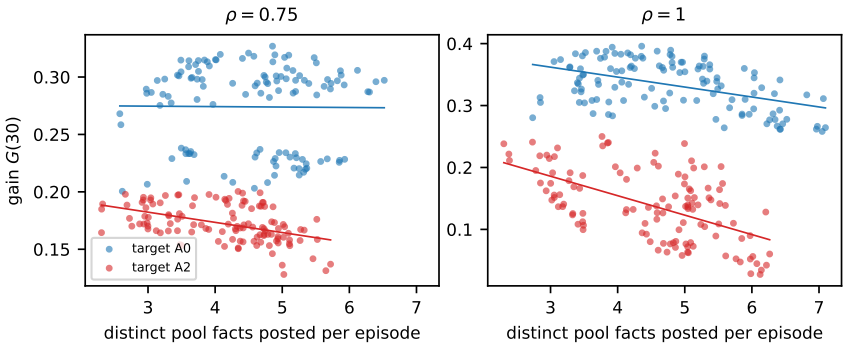


Figure 4: Each point is one setting (q, q_c, λ_c, gate): gain at $t = 30$ against the number of different pool facts the controller posted per episode. Lines: least-squares fits.

Why: saying more of the pool pulls toward the same point. Every fact is true, and the agents' facts plus the whole pool give (0.75, 0, 0.25). With full memory nothing posted is forgotten, so the more *different* pool facts a controller puts into circulation, the closer beliefs move to that point, whatever its target: down from the false controller's side, and down from the truth controller's best, which lies above 0.75. At $\rho = 1$ each additional distinct fact lowers the gain by 0.031 (false) and 0.016 (truth) (Figure 4, right). The rate sets that number: spread over 30 time units, the controller reads varied boards and posts more different facts (6.3 per episode at $\lambda_c = 1$, gate off); at $\lambda_c = 8$ it spends its budget in a burst on similar boards and repeats itself (5.0), which is why the gain partly recovers there. When agents forget, facts leave again and the effect is weak (Figure 4, left). This was found after looking at the data, so it was rerun with fresh random draws: **every one of the 16 checked reversals reappeared** (false controller, gain at $\lambda_c = 0.5 \ldots 8$: 0.20, 0.15, 0.08, 0.05, 0.05). The explanation by distinct facts rests on a correlation across settings and is not yet tested directly (for example with a controller that remembers what it has said).

4 The vote gate

The gate keeps the controller silent when it reads enough support. Turning it off adds about 16 messages per episode at $\lambda_c = 1$ (confirmed with fresh draws) and **no control** when agents forget (-0.017 and -0.003); with full memory it *lowers* the gain (truth -0.05 on average) through the mechanism above.

5 The controller overestimates its support

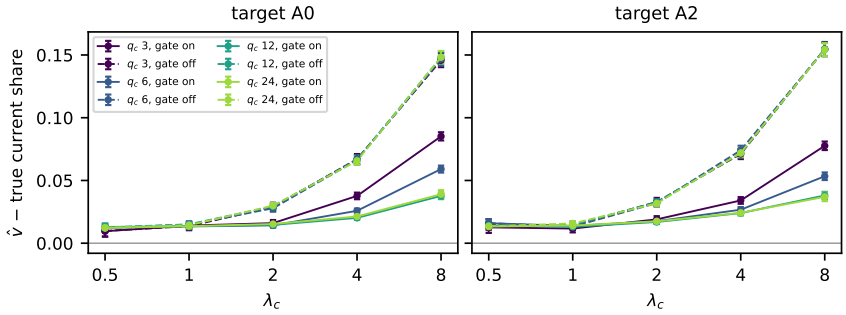


Figure 5: Error of the controller's estimate $\hat{v}$ (share of posts read whose author voted for the target) against the true current share of all 24 agents' last votes; $\rho = 0.75$; solid gate on, dashed off.

Unlike in Simulation 1, $\hat{v}$ is biased upward, and **the faster the controller, the larger the bias**: about $+0.014$ at $\lambda_c \leq 1$, rising to $+0.05$ (gate on) and $+0.15$ (gate off) at $\lambda_c = 8$. A likely reason, not yet tested: the posts it reads come from agents who acted recently, often just after reading the controller's facts, while the true share also counts older votes. More posts read (q_c) reduce the error's spread (0.16 at $q_c = 3$, 0.07 at $q_c = 24$) but not the bias.

6 How efficiently does the controller steer?

A controller can change the swarm in two ways: in the direction it is asked to, or in ways that happen whatever its target. We measure how readable the command is in the swarm, how far each controller gets toward its target, how much of its change is directed, what each fact it posts buys, and how unsure the agents become. Throughout, the target is chosen by a fair coin, and all quantities are taken at one time t, across the 1,000 episodes of each arm (not over whole trajectories). Setting: probability matching, $q = 6$.

6.1 Measures

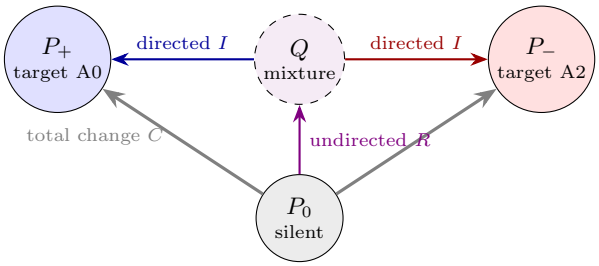


Diagram 2. Three distributions of the swarm's state at time t, one per arm, each over the episodes of that arm. The controller's total change C splits exactly into a part that depends on its target (I, from the mixture Q to each arm) and a part that does not (R, from silence to Q). Efficiency $\eta = I/C$ is the directed share.

Distributions. For each arm a (silent 0, target A0 $+$, target A2 $-$) and a time t, each episode e contributes a probability vector $y_a(e)$ over the values x of the state X: at *one-agent* level its vote shares (the vote of a randomly chosen agent), at *swarm* level a mark on the bin of $s = m_{\mathrm{A0}} - m_{\mathrm{A2}}$ (difference of mean beliefs, three equal bins on $[-1, 1]$). Then $\hat{P}_a = \frac{1}{n}\sum_e y_a(e)$. Any pair of controllers can be written as

$$\hat{P}_\pm = \hat{P}_0 + c \pm \Delta, \qquad c = \tfrac{1}{2}(\hat{P}_+ + \hat{P}_-) - \hat{P}_0, \quad \Delta = \tfrac{1}{2}(\hat{P}_+ - \hat{P}_-), \tag{2}$$

a *common* part c (what control does on average, whatever the target) and a *directional* part Δ.
Readability and the pair efficiency. With $\hat{Q} = \hat{P}_0 + c$ and $\mathrm{KL}(p\|q) = \sum_x p(x)\log_2[p(x)/q(x)]$,

$$I = \tfrac{1}{2}\mathrm{KL}(\hat{P}_+\|\hat{Q}) + \tfrac{1}{2}\mathrm{KL}(\hat{P}_-\|\hat{Q}), \quad C = \tfrac{1}{2}\mathrm{KL}(\hat{P}_+\|\hat{P}_0) + \tfrac{1}{2}\mathrm{KL}(\hat{P}_-\|\hat{P}_0) = I + R, \quad R = \mathrm{KL}(\hat{Q}\|\hat{P}_0), \quad \eta = I/C. \tag{3}$$

I is the mutual information between the target and the state: how readable the command is (at most 1 bit). I depends only on Δ, R only on c; the reference $\hat{P}_0$ is the silent arm. For small changes, $\mathrm{KL}(p + \epsilon\|p) \approx \sum_x \epsilon(x)^2/(2\ln 2\, p(x))$ gives

$$\eta \approx \frac{1}{1 + \|c\|^2/\|\Delta\|^2}, \qquad \|v\|^2 = \sum_x \frac{v(x)^2}{\hat{P}_0(x)}. \tag{4}$$

So $\eta = 1$ exactly when the two controllers' changes are mirror images ($c = 0$), whatever their size, and η falls only through the *ratio* of common to directional change: when both push the same way (a common push), or one moves the swarm much more than the other (a lopsided pair: along one direction with sizes a and b, $\|c\|/\|\Delta\| = |a-b|/(a+b)$). It is a property of the pair, not of either controller, and it stays constant whenever both effects grow together. At swarm level C is often infinite (the controlled swarm reaches states silence never visits); there we use $\eta_{JS} = I/(I + \mathrm{JS}(\hat{Q}, \hat{P}_0))$ with the bounded Jensen–Shannon divergence.

Each controller against its own target z (one agent, votes):

$$S_z = \mathrm{KL}\big(\mathrm{Bern}(\hat{P}_z(z)) \,\|\, \mathrm{Bern}(\hat{P}_0(z))\big), \qquad \pi_z = \frac{S_z}{\log_2[1/\hat{P}_0(z)]}, \qquad \varphi_z = \frac{S_z}{\mathrm{KL}(\hat{P}_z \| \hat{P}_0)}, \qquad \text{price} = \frac{S_z}{\text{facts posted}}. \tag{5}$$

S_z is the least information any intervention must inject to raise the target's share from its silent value to the observed one (raise the target, scale the other answers in proportion). **Progress** π_z is the share of the way from silence to unanimity on the target, in bits; **focus** φ_z the share of the controller's actual change that was needed for that gain; both are negative when the controller moved its target *down*. The **price** is the information injected per fact the controller posted. Finally, the agents' own uncertainty: $\Delta\bar{H}$ (mean entropy of their beliefs) and ΔJSD (disagreement, the entropy of the mean belief minus $\bar{H}$), controlled minus silent, same episode.

6.2 Over time, at one setting

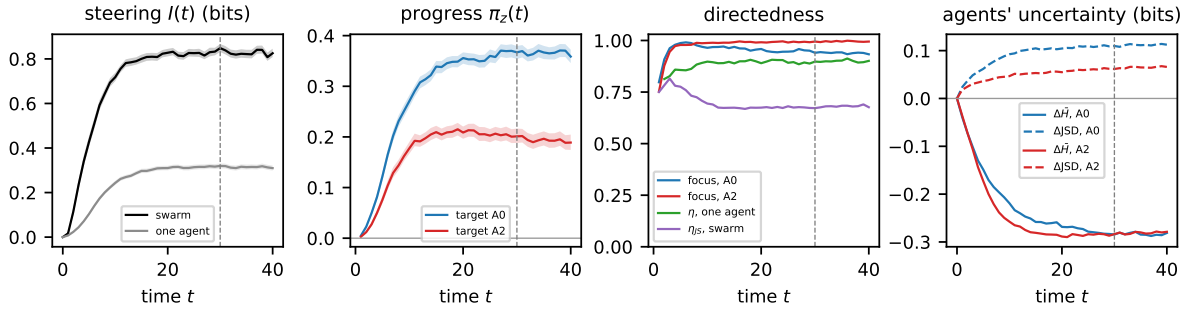


Figure 6: At $\rho = 0.75$, $q_c = 12$, $\lambda_c = 1$, gate on (the controller stops at $t = 30$, dashed): readability $I(t)$, each controller's progress $\pi_z(t)$, directedness (focus φ_z, pair efficiency η, η_{JS}), and the agents' uncertainty. Bands: 95% intervals over episodes.

- **Control builds within 10–15 time units and persists**: $\pi_{\mathrm{A0}} = 0.37$, $\pi_{\mathrm{A2}} = 0.20$ at $t = 30$, almost unchanged at $t = 40$; focus 0.94 and 0.99, $\eta = 0.90$.
- **Agents become surer** ($\Delta\bar{H} = -0.28$ bits) **but disagree more** (ΔJSD +0.11 for target A0, +0.06 for target A2).

6.3 Across the control plane

- **When agents forget, progress hardly depends on the rate, gate or** q_c (π_{A0} 0.28–0.39, π_{A2} 0.14–0.23), but the price does: up to 0.067 bits per fact at $\lambda_c = 0.5$ with the gate on, 0.016 at $\lambda_c = 8$; the gate makes slow control up to twice as cheap.
- **With full memory the false controller stalls as the rate rises**: π_{A2} falls from 0.11–0.18 at $\lambda_c = 0.5$ to 0.01–0.08 at $\lambda_c = 8$ (gate on) while π_{A0} stays at 0.30–0.49. Its focus stays near 1: it still pushes toward A2, it just gets almost nowhere (Section 3: more different pool facts pull both targets toward the full-pool belief).

6.4 Readability, pair efficiency and uncertainty against the controller rate

- **When agents forget,** η **is flat (0.86–0.95)**: the two controllers' effects grow together.
- **With full memory** η **falls with the rate** (0.93 to 0.67) because the pair turns lopsided: the false controller's progress collapses, the truth controller's does not. Swarm η_{JS} moves the other way, because swarm readability is saturated at 1 bit; the one-agent η is the informative one here.

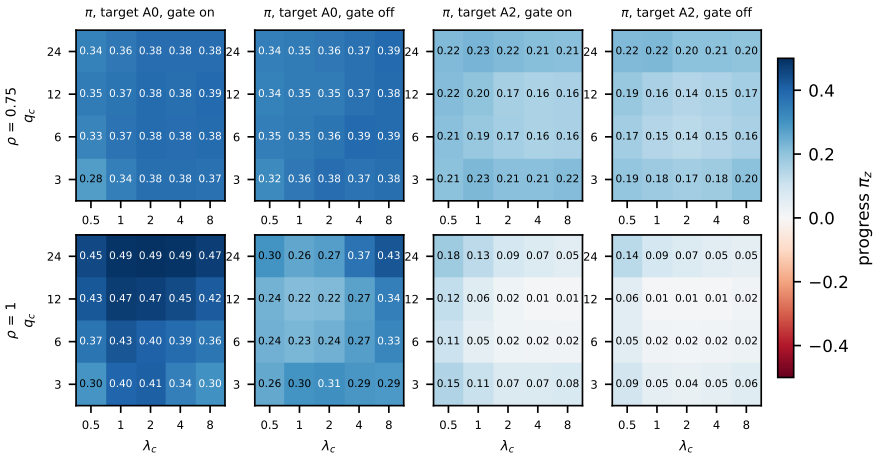


Figure 7: Progress π_z of each controller at $t = 30$ over the reading budget q_c and the controller rate λ_c (one agent, votes). Red: the controller moved its target down. Dash: undefined (silence already at unanimity).

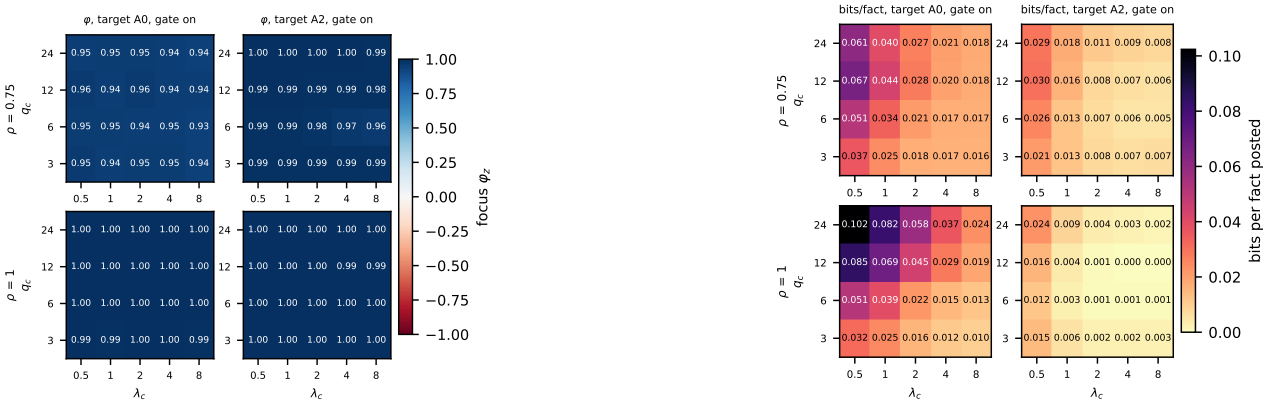


Figure 8: Left: focus φ_z (negative: the change went against the target). Right: price, information injected per fact posted (bits). $t = 30$; gate on. Dash: undefined (no change, or the controlled swarm considers an answer silence rules out completely).

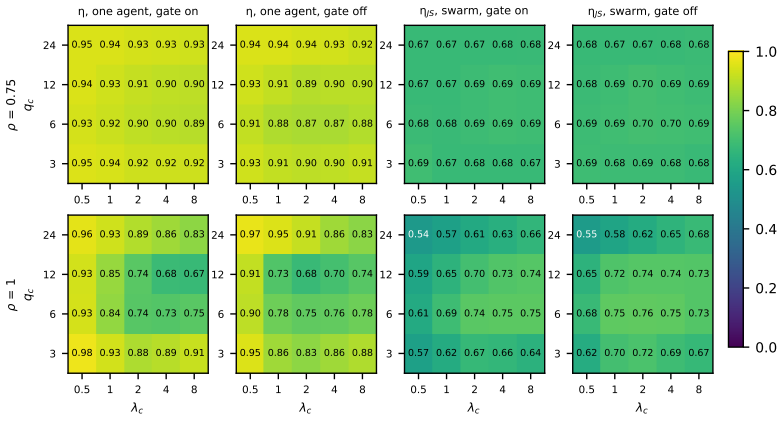


Figure 9: Pair efficiency at $t = 30$ over q_c and the controller rate λ_c: one-agent η (beliefs) and swarm η_{JS}. Dash: undefined.

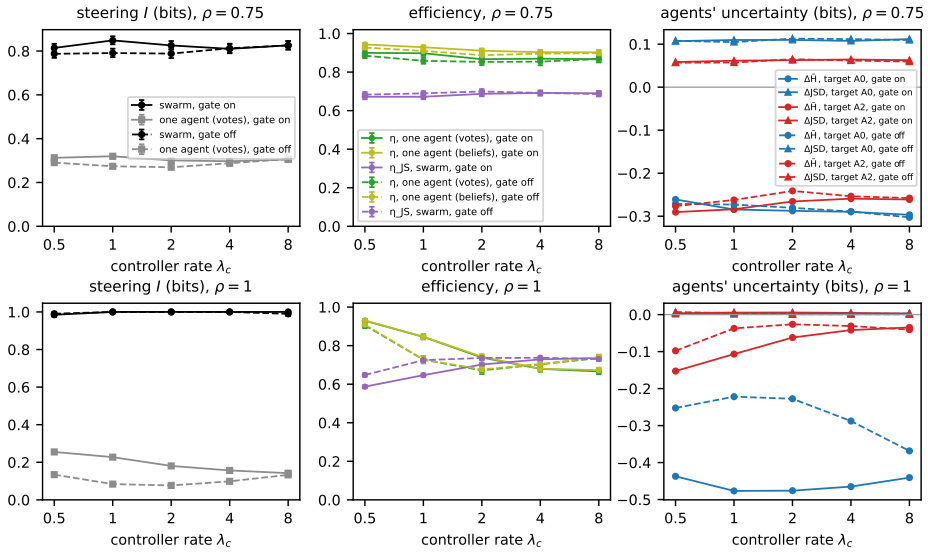


Figure 10: Readability I (left), pair efficiency (middle) and the change in the agents' uncertainty (right; circles $\Delta\bar{H}$, triangles ΔJSD) at $t = 30$ against the controller rate λ_c ($q_c = 12$; solid: gate on; dashed: gate off). Bars: 95% intervals.

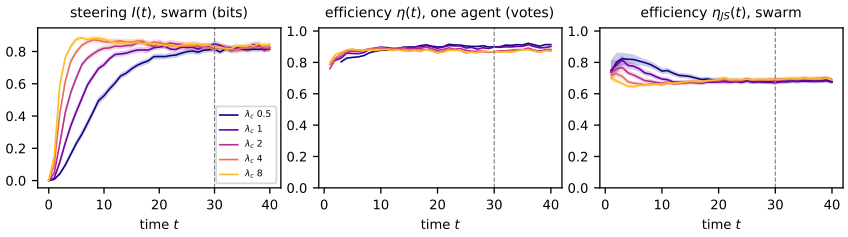


Figure 11: Readability and pair efficiency over time, $\rho = 0.75$, $q_c = 12$, gate on, one line per rate; the controller stops at $t = 30$ (dashed). Bands: 95% intervals.

6.5 Precision at 50, 100 and 300 paths

Table 2: Value from 1,000 paths per arm, and root-mean-square error of the estimate from 50, 100 or 300 paths (200 random subsets each), at $t = 30$, $\lambda_c = 1$, $q_c = 12$, gate on, $\rho = 0.75$. *: share of subsets in which the quantity is infinite or undefined.

quantity	1,000 paths	error, 50	error, 100	error, 300
steering I, swarm (3 bins)	0.849	0.039	0.026	0.013
steering I, one agent (votes)	0.320	0.023	0.017	0.009
efficiency η, one agent (votes)	0.898	0.032	0.022	0.010
efficiency η_{JS}, swarm	0.672	0.027	0.019	0.010
progress π, target A0 (votes)	0.367	0.029	0.018	0.009
progress π, target A2 (votes)	0.201	0.031	0.022	0.011
focus φ, target A0 (votes)	0.940	0.022	0.016	0.009
focus φ, target A2 (votes)	0.995	0.013	0.007	0.004
$\Delta\bar{H}$, target A0	-0.284	0.024	0.016	0.009
$\Delta\bar{H}$, target A2	-0.284	0.023	0.016	0.008
ΔJSD, target A0	0.110	0.008	0.006	0.003
ΔJSD, target A2	0.062	0.008	0.006	0.003

Across all settings, at 100 paths: readability ±0.01 (one agent) to ±0.03 (swarm), progress ±0.02, focus ±0.01–0.02, η ±0.02–0.03, $\Delta\bar{H}$ ±0.02, ΔJSD ±0.006. Two settings that differ by at least 0.02 are ordered correctly in 90–100% of subsets.

Caveats. One world (task_003), 24 agents, exact-Bayesian agents, probability-matching votes; a controller without memory that posts one fact at a time; a budget of 30 messages. Means over 1,000 episodes per setting. Full analysis, code and tables: `analysis/task003_llm_free/simulation_2/analysis/` (report of 9 October, reviewer guide).